Do not write outside the box

0 1 The Earth's atmosphere has changed during the last 4.6 billion years.

0 1 . 1 What is the approximate percentage of nitrogen and of oxygen in the Earth's atmosphere today?

Draw **one** line from each gas to the percentage of that gas.

[2 marks]

Gas	**Percentage (%) of gas**
	20
Nitrogen	40
Oxygen	60
	80

0 1 . 2 The approximate percentage of carbon dioxide in the Earth's early atmosphere was 95%.

Which are **two** reasons why the percentage of carbon dioxide has **decreased** since the Earth's early atmosphere?

[2 marks]

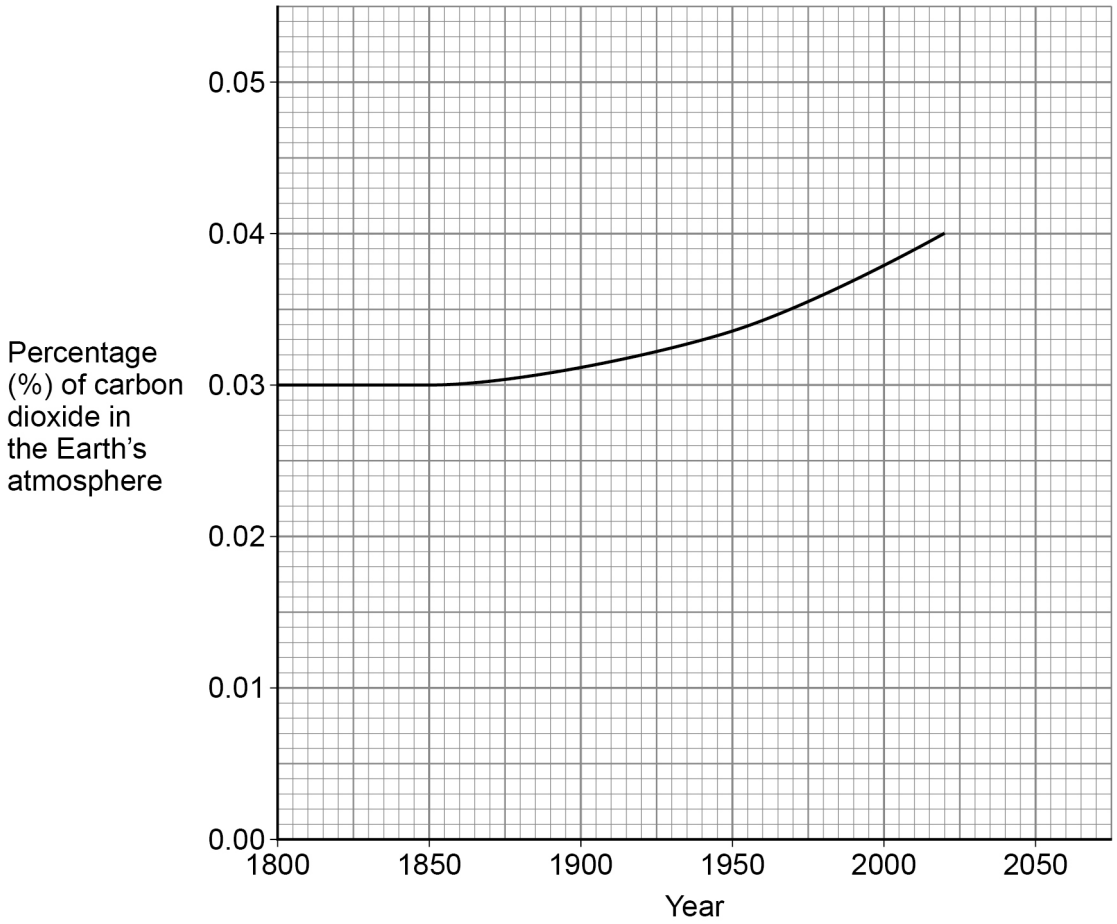
Tick (✓) **two** boxes.

- Combustion of fuels ☐
- Deforestation ☐
- Dissolving in oceans ☐
- Photosynthesis ☐
- Respiration ☐

Do not write outside the box

0 1 . 3 **Figure 1** shows the change in the percentage of carbon dioxide in the Earth's atmosphere from 1800 to 2020.

Figure 1



Describe the trend in the percentage of carbon dioxide in the Earth's atmosphere from 1800 to 2020.

Use data from **Figure 1**.

[3 marks]

Do not write outside the box

Carbon dioxide is a greenhouse gas.

0 1 . 4 Which of the following is also a greenhouse gas?

[1 mark]

Tick (✓) **one** box.

Argon ☐

Methane ☐

Nitrogen ☐

Oxygen ☐

0 1 . 5 Which of the following is an environmental problem caused by greenhouse gases?

[1 mark]

Tick (✓) **one** box.

Acid rain ☐

Climate change ☐

Global dimming ☐

Do not write outside the box

0 1 . 6 Calculate the relative formula mass (M_r) of carbon dioxide (CO_2).

Relative atomic masses (A_r): C = 12 O = 16

[2 marks]

_____

_____

_____

Relative formula mass of carbon dioxide = _____

11

Turn over for the next question